

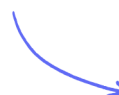
Multiple Choice Questions

Formulae & Equations

Formulae & Mass / Avogadro & the Mole / Full & Ionic Equations / Calculating Formulae

Easy (3 questions)	/3
Medium (6 questions)	/8
Total Marks	/11

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Easy Questions

1 Which of these compounds has the same empirical and molecular formulae?

- A. C_2H_4
- B. C_3H_8
- C. C_4H_{10}
- D. C_5H_{10}

(1 mark)

2 Which of these atoms has the most neutrons?

- A. $^{115}_{49}\text{In}$
- B. $^{124}_{50}\text{Sn}$
- C. $^{123}_{51}\text{Sb}$
- D. $^{124}_{52}\text{Te}$

(1 mark)

3 What is the relative formula mass of hydrated sodium carbonate, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$?

[A_r values: H = 1.0 C = 12.0 O = 16.0 Na = 23.0]

- A. 106
- B. 142
- C. 263
- D. 286

(1 mark)

Medium Questions

1 There are 6.02×10^{23} atoms in 0.25 mol of

- A. He
- B. H_2O
- C. BH_3
- D. CH_4

(1 mark)

- 2 (a)** Ammonium iron(II) sulfate, $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$, is a double salt that is used as a source of iron(II) ions.

What is the relative formula mass of the double salt?

[A_r values: H = 1.0 N = 14.0 O = 16.0 S = 32.1 Fe = 55.8]

- A.** 277.9
- B.** 284.0
- C.** 392.0
- D.** 447.8

(1 mark)

- (b)** Ammonium sulfate is used in the preparation of the double salt.

What types of bond are present in ammonium sulfate?

- A.** ionic only
- B.** covalent and ionic only
- C.** dative covalent and ionic only
- D.** ionic, covalent and dative covalent

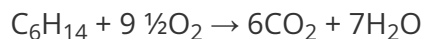
(1 mark)

- (c)** What is the **total** number of ions present in 0.1 mol of the double salt? [Avogadro constant (L) = $6.02 \times 10^{23} \text{mol}^{-1}$]

- A.** 1.80×10^{23}
- B.** 2.41×10^{23}
- C.** 3.01×10^{23}
- D.** 6.62×10^{23}

(1 mark)

3 The equation for the complete combustion of hexane is shown.



How many molecules of carbon dioxide are formed when 2×10^{-3} mol of hexane undergoes complete combustion? [Avogadro constant $L = 6.02 \times 10^{23} \text{ mol}^{-1}$]

- A. 1.20×10^{21}
- B. 7.22×10^{21}
- C. 8.43×10^{21}
- D. 3.18×10^{23}

(1 mark)

4 Barium hydroxide reacts with sulfuric acid as shown.



Which is the ionic equation for this reaction?

- A. $\text{Ba}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) + 2\text{H}^{+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l})$
- B. $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$
- C. $\text{OH}^{-}(\text{aq}) + \text{H}^{+}(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$
- D. $\text{Ba}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) + 2\text{H}^{+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{Ba}^{2+}(\text{s}) + \text{SO}_4^{2-}(\text{s}) + 2\text{H}_2\text{O}(\text{l})$

(1 mark)

5 An oxide of lead contains 90.7% by mass of lead.

What is the formula of this oxide?

[A_r values: O = 16.0 Pb = 207.2]

- A. PbO
- B. PbO₂
- C. Pb₂O₃
- D. Pb₃O₄

(1 mark)

6 A sample of seawater with a mass of 1 kg contains 6×10^{-9} g of gold.

How many atoms of gold, to one significant figure, are there in 1 g of this seawater?

[A_r value: Au = 197 Avogadro constant = $6 \times 10^{23} \text{ mol}^{-1}$]

A. 2×10^{10}

B. 4×10^{12}

C. 2×10^{13}

D. 4×10^{15}

(1 mark)